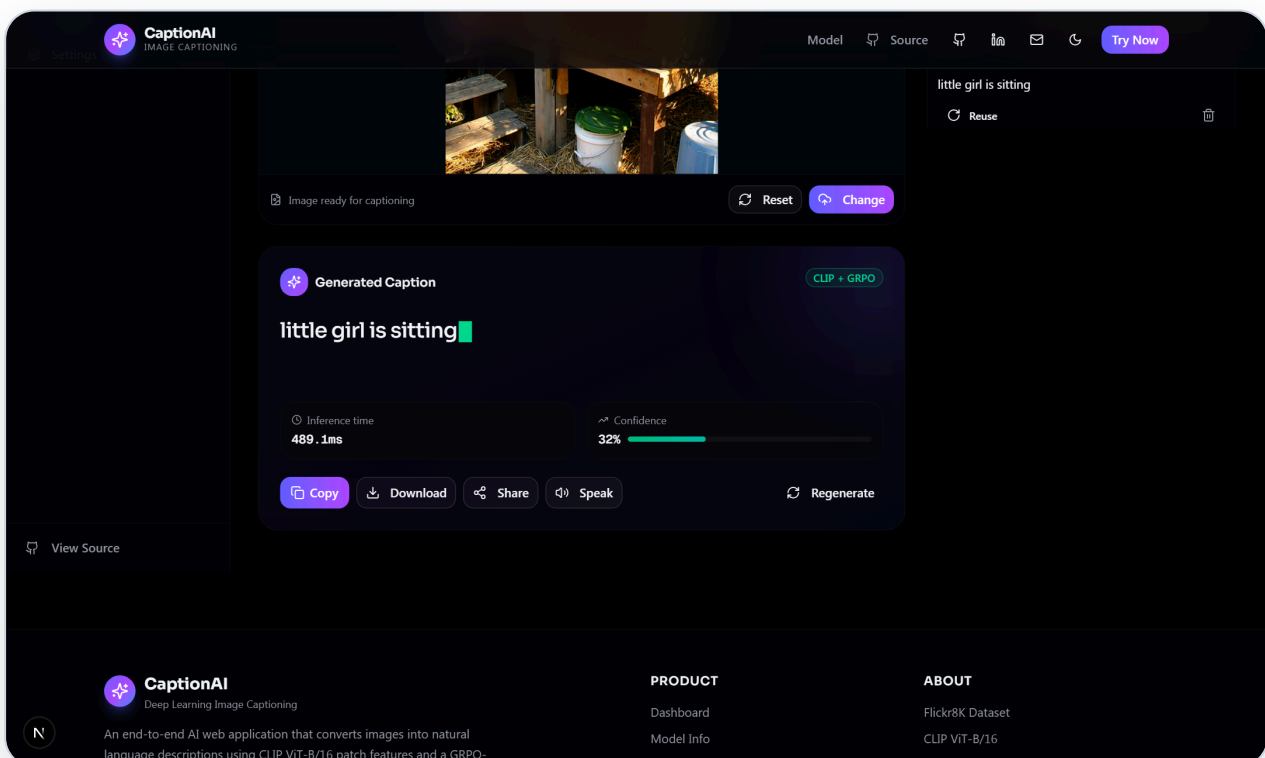


CaptionAI

A web app that looks at a photo and describes it in plain English

In one paragraph: CaptionAI is a website where you drop in any picture — from your phone, camera or screen — and within about half a second it writes a sentence describing what it sees: *"little girl is sitting."* It also tells you how sure it is, and when a picture is outside its comfort zone (a cartoon, a screenshot), it automatically hands the job to a second, more general model instead of guessing badly. It is built for teachers and presenters who want to show what modern AI can do live in front of a room, and for students and researchers who want to see exactly how such a system is built, trained and measured — every number and experiment behind it is documented inside the app itself.



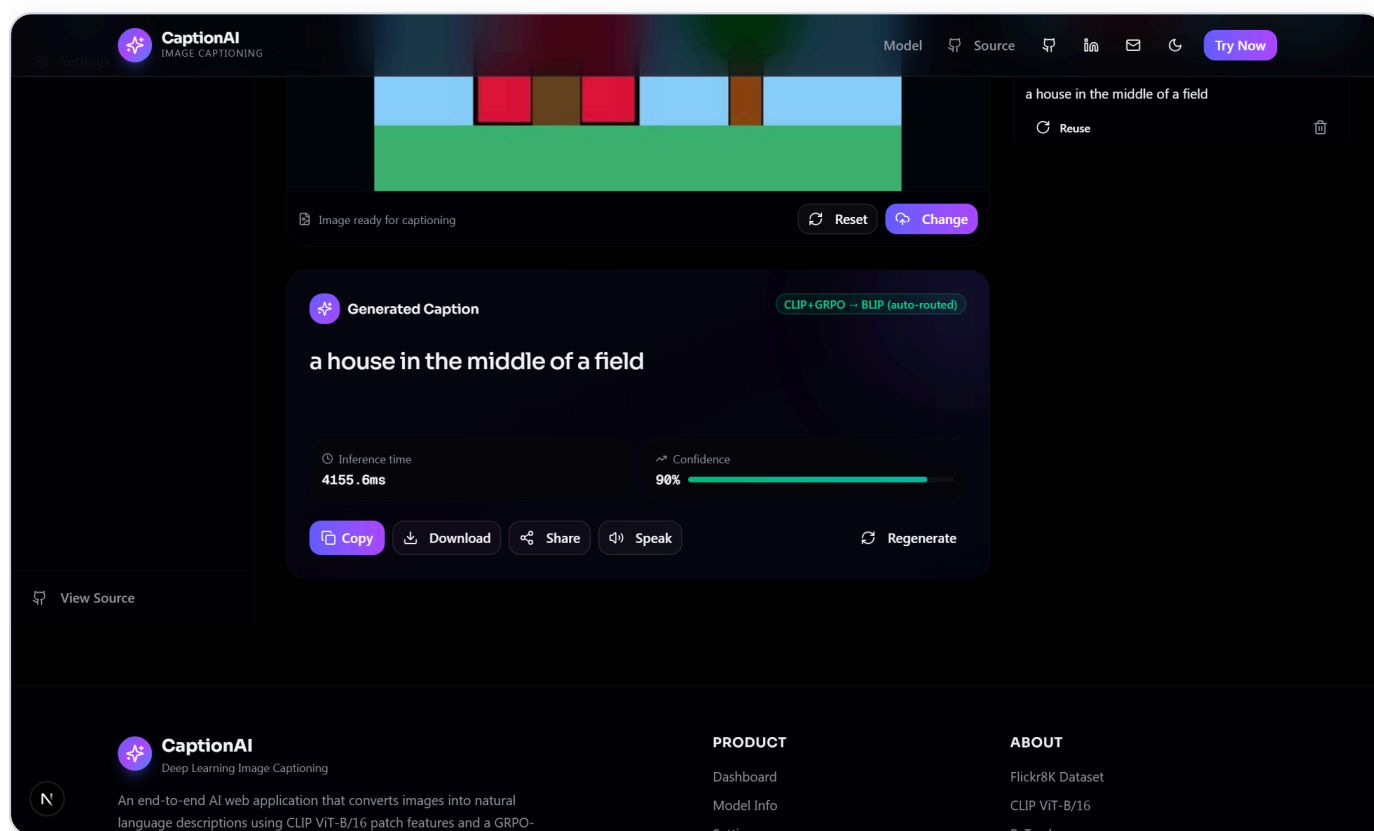
The live app captioning a real photo — result card shows the caption, the model that answered, how sure it is, and how long it took.

What you can do with it — by who you are

Teacher / Presenter

Demo AI live, with nothing to prepare

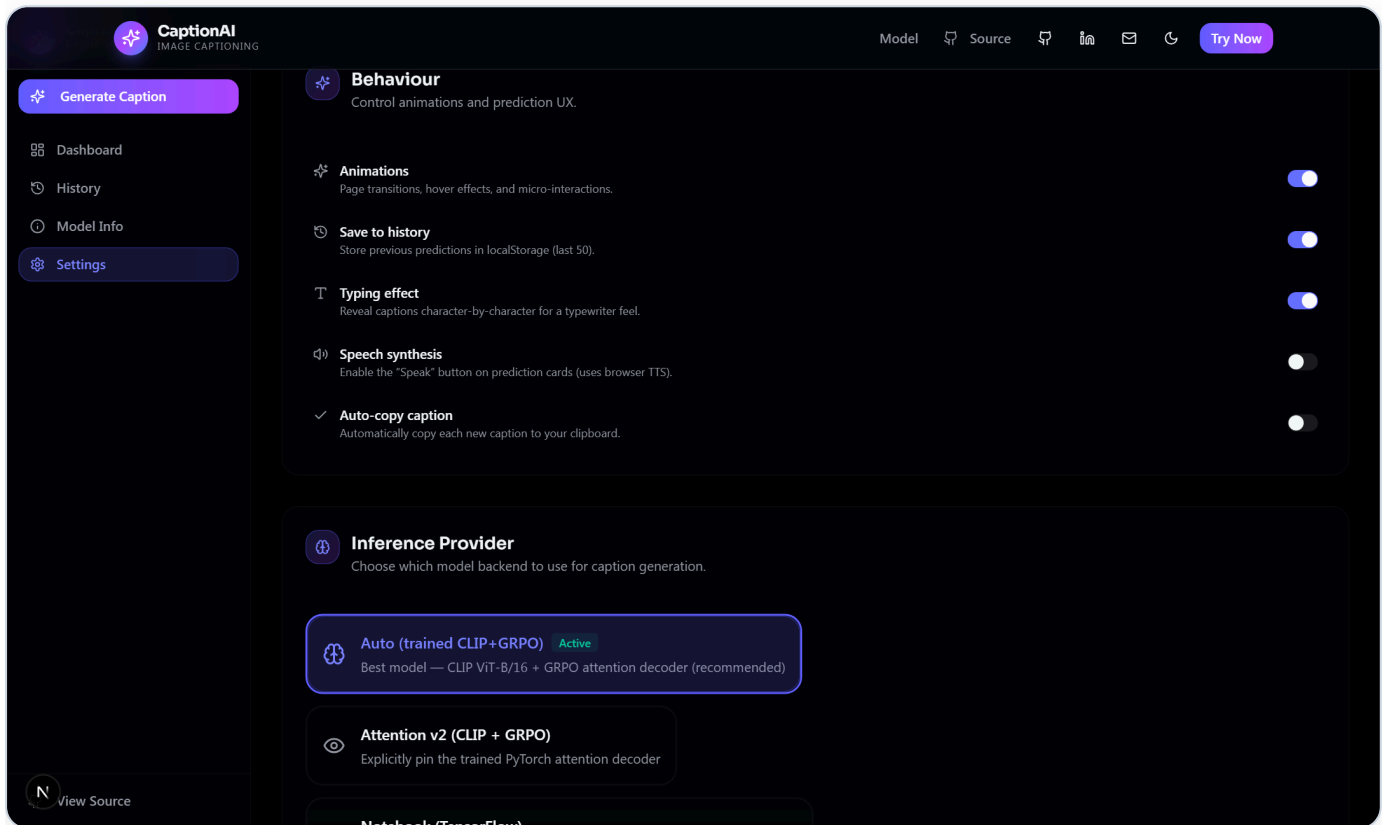
- **Show a caption in real time.** Upload any classroom-relevant photo during the talk and watch the sentence appear in about half a second — no waiting, no slides.
- **Demonstrate AI confidence honestly.** Every result shows a certainty meter; captions can be copied, downloaded, or read aloud on the spot.
- **Make the “AI gets confused” point visually.** Drop in a cartoon or a phone screenshot and the badge on the result changes to show it was automatically routed to the general model — a live lesson about where AI is strong and where it hands off.



A cartoon house captioned “a house in the middle of a field” — note the “auto-routed” badge: the app detected an unusual image and asked the general model instead of guessing.

A complete, honest record of how this AI was built

- **Read the whole journey in the app.** The Model Info page walks through every version of the brain inside — from the first rough attempt to the final model, with the exact scores at each step.
- **Learn from the failures, not just the wins.** Twenty techniques are documented as expandable cards: ten that worked and ten that didn't, each with the measurements that decided it.
- **Compare models yourself.** Settings lets you switch which engine answers — the fast trained model, the general fallback, or the original first attempt — and see the difference on the same photo.



The settings screen: pick which model answers, or let the app decide automatically (recommended).

Fast captions, no account, nothing stored

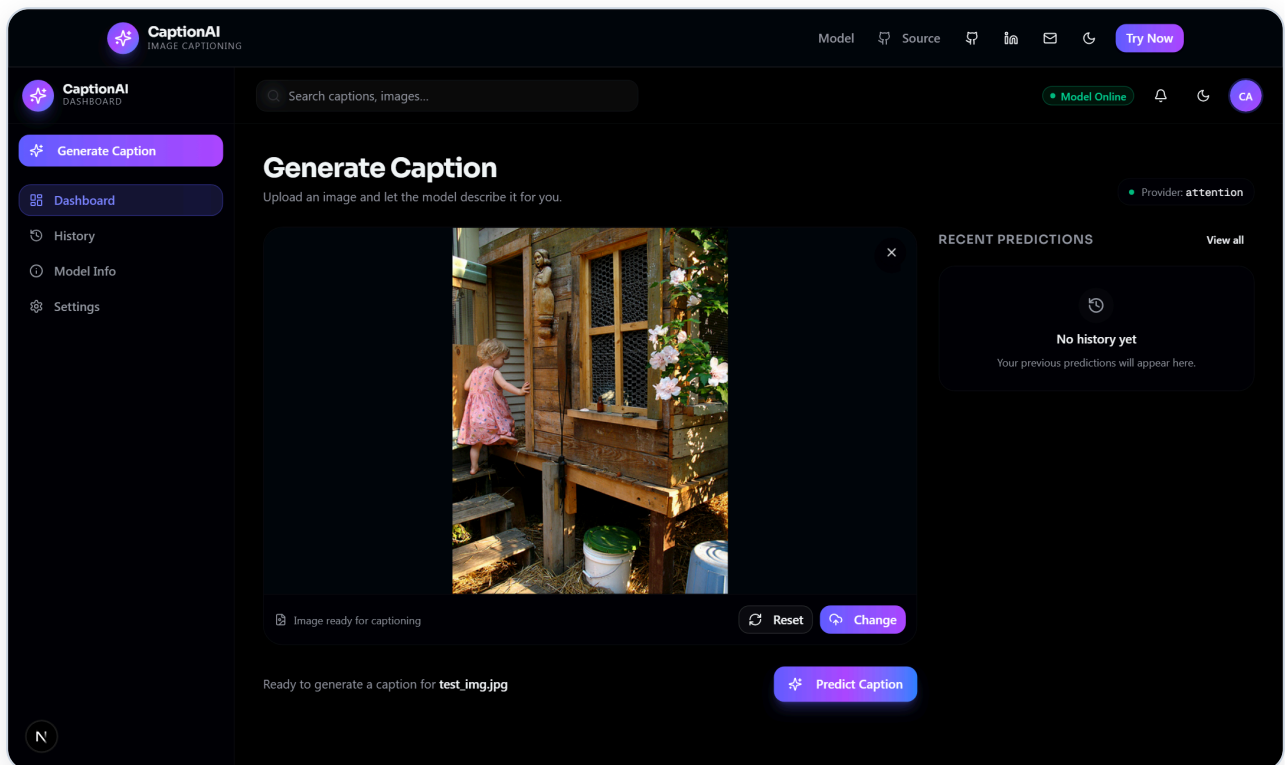
- Open the page, drop an image, read the caption. Copy it, download it, or have it spoken aloud.
- Your recent captions appear beside you for reuse — they are saved only inside your own computer's browser (the last 50), never on a server.

Requirements checklist — in plain language

What was asked for	Done?	What that means for you
Improve the accuracy of the trained caption model	Yes	The model now describes photos roughly 23% better than the starting point, and it catches its own mistakes (mentions fewer objects that aren't in the picture).
Document what improved and what didn't, with real numbers	Yes	A full research report plus an in-app Techniques page — every experiment, every number, including the ones that failed. Nothing is hidden or rounded up.
A working web app: upload → caption, clean and fast	Yes	About half a second per caption on an ordinary laptop, no graphics card needed. Crashes that used to freeze the machine during development were fully fixed.
Handle images the model wasn't trained on (cartoons, screenshots, food shots)	Yes	Built-in auto-detection hands those to the general model. Out of 10 unusual test images, all 10 were caught correctly.
Confidence scores users can trust	Yes	The certainty meter was recalibrated — it now matches how often the model is actually right (before, it showed inflated confidence).
Run it online so other people can try it	Not yet	This is the one open item: today the app runs on the developer's own computer. Making it a public link is a ready-to-execute step — we documented three free/cheap hosting paths — and needs your go-ahead to pick one.
Automatic quality checks for future updates	Next phase	Everything in this version was verified by hand across every screen and image type. The next phase adds automated checks so future changes can be verified without re-testing everything manually.

Quick start — six steps from zero to demo

- 1 **Open the app.** You land on the intro page; press **Try Now** (or the dashboard link) to enter the working screen. Everything runs from one address — no install, no login.
- 2 **Drop a photo** into the upload box, or press the box and choose a file. You'll see a preview and an "image ready" confirmation.



- 3 **Press "Predict Caption."** About half a second later the result card appears: the sentence, the model badge, the certainty meter, and how long it took. Use **Copy**, **Download** or **Speak** beside it.
- 4 **Try a cartoon or screenshot** to show the smart hand-off: the badge on the result changes to "auto-routed," proving the system noticed the image is outside its specialty and asked the general model — take a few seconds longer, but far more suitable for odd inputs.
- 5 **Open Model Info** from the sidebar to show the story behind the AI — the champion model's scorecard, the generation history, and the twenty technique cards that expand to show what was tried, what worked, and what didn't.
- 6 **Check Settings** to see (or switch) which engine answers. "Auto" with the trained model is the recommended choice; the other engines are there for comparison and teaching.

What's next / open items

- **Put it online.** Recommendation: start with a free tunnel from the current machine for demos this week, then move to a free/low-cost always-on host for a permanent public link. We've already researched the options — this is a decision plus about an hour of setup, not new development.
- **Automated quality checks.** So any future change to the app can be verified in minutes instead of a manual sweep of every screen. A contained, low-risk next phase.
- **Richer report (optional).** The research document already has the full story; adding a few more industry-standard scoring methods would make the comparisons even more complete — valuable only if this goes to academic reviewers.

Support & data safety

There are no accounts and no database in this version: photos you test are processed and answered on the spot — nothing is uploaded to a third party, nothing about you is stored on any server, and your recent-captions list lives only inside your own computer's browser. The app's "password" layer doesn't exist because there's nothing to protect — anyone with the link can use the captioner, which is exactly what you want for demos (and the reason to keep the link to yourself until you decide it should be public). Backups are straightforward: the entire project, including the trained model, lives in one folder that is version-controlled and mirrored to cloud storage, so any change can be rolled back. If something stops responding, the fix is a single restart of the app on the host machine — the same way you restart any web service — and the health check in the top bar shows green when it's up and reachable. For anything else, contact the developer, **Muhammad Hamza Mushtaq**.